MIL-M-25393A(USAF)
1 July 1969
SUPERSEDING
MIL-H-25393(USAF)
20 September 1955

MILITARY SPECIFICATION
MANUAL, TECHNICAL: INTERMEDIATE MAINTENANCE INSTRUCTIONS
(Airborne Equipment)

1. SCOPE

1.1 This specification prescribes the requirements for the preparation of technical manuals of intermediate maintenance instructions for airborne (guided missile or aircraft) equipment of a mechanical, electrical, pneumatic, or hydraulic nature, or combinations thereof: for example; complex accessories, instruments, propellers, photographic equipment, armament equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-M-38730	Manuals, Technical: General Requirements for Preparation of
MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

STANDARDS

Military

MIL-STD-838	Lubrication of Military Equipment

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 275	Guide for the Use of Lubricants, Compounds and Fluids in Aircraft

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preparation. The general manner of preparation of text, illustrations, diagrams, negatives, etc, shall be in accordance with MIL-M-38784 and MIL-P-38790. The manual shall contain all information required by intermediate maintenance organizations, with the tools and test equipment authorized. The manual shall not cover maintenance which requires tools, test equipment, manpower, skills, etc, for overhaul, or authorized for depot organizations only.

3.2 Arrangement. The manual shall be arranged as follows:

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Front Matter

Title Page

List of Effective Pages

Table of Contents

List of Illustrations

List of Tables

Introduction

Section I, Description

Section II, Special Tools and Test Equipment

Section III, Preparation for Maintenance

Section IV, Check-Out and Analysis

Section V, Repair Procedures

Section VI, Difference Data Sheets

3.3 Front matter. The title page, list of effective pages, table of contents, list of illustrations and list of tables, shall be prepared in accordance with the requirements of MIL-M-38784.

3.3.1 Introduction. The requirements of MIL-M-38784 are applicable. In addition, when repair parts for the equipment are to be supplied in the form of kits, the introduction shall contain the following statement: "Some repair parts for the equipment covered in this manual are provided in the form of kits. Refer to the applicable Illustrated Parts Breakdown for detailed information on the contents of the kits. Activities shall replace all parts, regardless of condition, which are removed in the process of disassembly with all like parts furnished in the kit. Therefore, instructions for cleaning, inspecting, and reworking such used parts have been omitted. If any parts in the kit must be cleaned, inspected, or tested prior to installation, instructions for performing these requirements are included in this manual. Naturally, all defective parts are to be replaced, but a part unnecessary to be removed in the process of disassembly shall not be removed solely for the purpose of replacement by a corresponding kitted part. Refer to AFLCR 65-42 regarding residue from kits and removed parts in this category."

3.4 Section I, Description. This section shall contain the following: the major characteristics, limitations, and leading particulars of the equipment; the purpose of the equipment and how it accomplishes its mission; a composite illustration of the separate components, citing nomenclature and type designations.

3.5 Section II, Special Tools and Test Equipment. This section shall list and illustrate the special tools and test equipment required for the work described. Such items shall be selected from the list of tools and test equipment approved by the Government by means of AGE Listings, Provisioning Conferences, Engineering Change Proposals (ECPs), Procurement Documents. Standard types of tools, such as screwdrivers, pliers, shall not be listed. Standard types of test equipment, such as voltmeters, tube testers, shall be listed. When a contractor cannot obtain an approved list, a manual shall show the contractor's recommended special tools and test equipment. However, these recommendations shall be changed promptly, if necessary, to conform to the official Government list as soon as such information is provided to the contractor.

3.5.1 Special tools list. The list of special tools shall be arranged in the following format. Tools not illustrated in use in other sections of the manual shall be illustrated following the list in this section.

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SPECIAL TOOLS LIST

Part (Tool) Number	Figure & Index No.	Nomenclature	Use

a. Part (tool) number. Tools shall be listed in sequence by part (tool) number. The part (tool) number shall agree with the drawing number.

b. Figure and index number. The figure and index numbers that appear in other sections, or this section, of the manual, shall be shown.

c. Nomenclature. The nomenclature shall agree with the drawing title.

d. Use. The purpose of each tool shall be stated. If the use is described in any part of the manual, reference to the applicable paragraph will suffice.

3.5.2 Test equipment list. The list of test equipment shall be arranged in the following format. Test equipment not illustrated in use in other sections of the manual shall be illustrated following the list in this section.

Type Designation	Alternate Type Designation	Figure & Index No.	Nomenclature	Use

a. Type designation. Test equipment shall be listed in alpha-numeric sequence by type designation, using the AN type designation, if assigned, or the commercial or manufacturer's designation if an AN type was not assigned.

b. Alternate type designation. If an alternate item of test equipment can be used, its AN type designation, if assigned, or the commercial or manufacturer's designation, if an AN type was not assigned, shall be listed.

c. Figure and index number. The figure and index numbers that appear in other sections, or this section, of the manual, shall be shown.

d. Nomenclature. The nomenclature shall agree with the drawing title.

e. Use. The purpose of each item of test equipment shall be stated. If the use is described in any part of the manual, reference to the applicable paragraph will suffice.

3.6 Section III, Preparation for Maintenance. This section shall describe and illustrate the procedures required to uncrate and unpack the equipment and set it up for intermediate maintenance. Instructions shall include precautions necessary to prevent damage.

3.7 Section IV, Check-Out and Analysis. This section shall provide instructions for determining whether the equipment meets minimum performance standards, and methods for identifying, locating, and remedying malfunctions. This shall include:

3.7.1 Bench test and performance standards. When applicable, illustrations shall be provided showing the recommended bench test setup, including intercomponent cabling and connections. Performance procedures shall be stated. One illustration of a setup will suffice when separate performance procedures utilize the identical setup. Test values and acceptable tolerances shall be given for each test.

3.7.2 Analysis. A troubleshooting guide shall be provided, identifying each malfunction, or symptom of malfunction; giving instructions for isolating the cause to a specific component and specifying the necessary adjustment, repair, or replacement.

3.8 Section V, Repair Procedures. This section shall contain instructions, illustrations and diagrams, for all disassembly, cleaning, inspection, repair, reassembly, and replacement involved in intermediate maintenance.

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3.8.1 Disassembly. Instructions for disassembly of components and accessories shall be only to the extent authorized for intermediate maintenance.

3.8.1.1 Kits. When repair parts for the equipment are to be supplied in the form of kits, the following note shall be placed at the conclusion of the disassembly instructions: "Presence of a new part in the applicable kit eliminates the necessity of cleaning, inspecting, or reworking the equivalent used part removed from the assembly being repaired. Refer to AFLCR 65-42 regarding residue from kits and removed parts in this category. The instructions which follow cover removed parts not supplied in kits and kitted parts (if any) which require cleaning, inspecting, or testing prior to installation."

3.8.2 Cleaning. Instructions and specifications shall be given for any cleaning methods to be used. Use of corrosion-preventive compounds on all steel parts after cleaning shall be stated. Information shall be provided for any kitted parts which must be cleaned prior to installation.

3.8.3 Inspection. Inspection of disassembled equipment for damage or wear shall be described. This information shall be in tabular or chart form, wherever practicable, with emphasis upon allowable service limits, wear, backlash, end play, balance, length and depth of scoring, etc. These tolerances are not to be confused with manufacturing tolerances; they are tolerances that may or may not exceed manufacturing tolerances, yet will not impair performance. This information shall be sufficiently comprehensive and adequate to serve as standards by which experienced technicians may determine when parts may be continued in use and when they must be replaced. Instructions for any kitted parts which must be inspected or tested prior to installation shall be given.

3.8.4 Repair. Detailed repair instructions shall be limited to those authorized for intermediate maintenance.

3.8.5 Lubrication. Lubrication instructions shall be those required for intermediate maintenance. The requirements of MIL-STD-838 and ANA Bulletin No. 275 are applicable.

3.8.6 Reassembly. Instructions and illustrations shall be provided for reassembly of subassemblies.

3.8.7 Final assembly. Instructions shall be given for final assembly of subassemblies to form the complete equipment. Reference shall be made to Section IV of the manual for testing the reassembled equipment.

3.9 Section VI. Difference Data Sheets. This section, when applicable, shall consist of Difference Data Sheets for additional models of the equipment. A separate Difference Data Sheet shall be prepared for each additional model to specify in the briefest practical form how intermediate maintenance for that model differs from the procedures given for the basic model in preceding sections.

3.9.1 Format. The requirements of MIL-M-38730 are applicable. The Difference Data Sheet(s) for a particular model shall start on a right-hand page. Page numbers shall run consecutively throughout the section. They shall be added chronologically as they are issued.

3.9.2 Content. Difference Data Sheets shall contain the following, as appropriate, to compress in one or two pages, and not more than four pages, the information on differences between the added model and the basic model covered in the preceding sections.

a. An external view of the model, if different than the one shown in Section I.

b. A table of leading particulars or a table of characteristic differences similar to that contained in Section I.

c. Illustrations which are essential for clarifying differences. Illustrations in preceding sections shall not be duplicated but referenced. Reference shall

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not be made to illustrations on other Difference Data Sheets.

d. Brief coverage of subjects specified for Sections II through V by appropriate statements such as "Disassembly. Same as Model No._____ except that .....", "Inspection. Same as Model No._____.", "Lubrication. None." Sentences shall be in short telegraphic style.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1 Intended use. The manuals prepared in accordance with this specification are intended for use by intermediate maintenance personnel.

Custodian:

Air Force - 16

Preparing Activity:

Air Force - 16

Review Activities:

Air Force - 01, 10, 70, 71, 80, 82, 84.

Project TMSS-F325

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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